



NORTH SYDNEY BOYS HIGH SCHOOL

2014 Year 11 Task 3

Mathematics Extension 1

General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lin
- ☐ Mr Weiss
- ☐ Mrs Ziazaris

Student Name: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	10	11	Total	Total %
Mark	$\frac{\quad}{5}$	$\frac{\quad}{5}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{50}$	$\frac{\quad}{100}$

Section I: Objective Response (5 marks)

Mark your answers for Questions 1 to 5 by shading the multiple choice box below :

1	Ⓐ	Ⓑ	Ⓒ	Ⓓ
2	Ⓐ	Ⓑ	Ⓒ	Ⓓ
3	Ⓐ	Ⓑ	Ⓒ	Ⓓ
4	Ⓐ	Ⓑ	Ⓒ	Ⓓ
5	Ⓐ	Ⓑ	Ⓒ	Ⓓ

Marks

- 1 Which value of m makes the points $(4, -3)$, $(0, m)$ and $(-2, 5)$ collinear?

1

(A) $m = 1$

(B) $m = \frac{7}{3}$

(C) $m = 4$

(D) $m = -\frac{1}{2}$

- 2 The value of $\lim_{x \rightarrow \infty} \frac{3x - 2x^2}{5x^2 + 2x - 1}$ is:

1

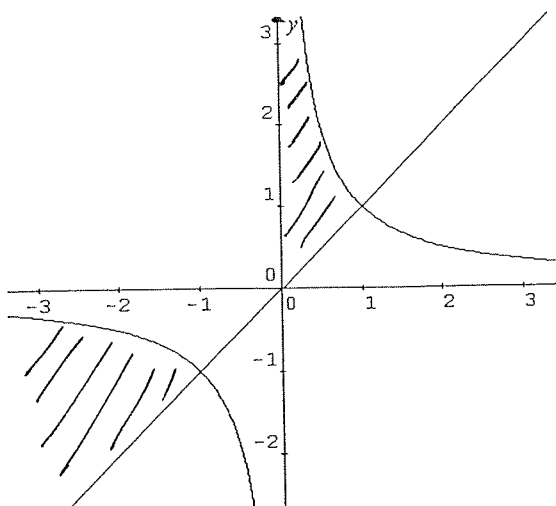
(A) $\frac{3}{2}$

(B) $\frac{3}{5}$

(C) $\frac{-2}{5}$

(D) $\frac{1}{2}$

3



The shaded area below represents which inequalities?

1

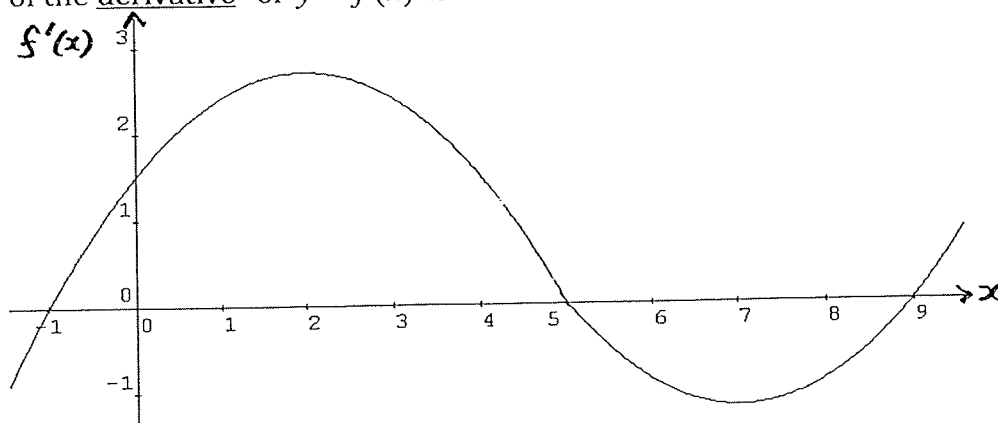
- (A) $y \leq \frac{1}{x}$ and $x - y \geq 0$
 (B) $y \leq \frac{1}{x}$ and $x - y \leq 0$
 (C) $y \geq \frac{1}{x}$ and $x - y \geq 0$
 (D) $y \geq \frac{1}{x}$ and $x - y \leq 0$

4 For what values of x is the curve $f(x) = x^3 + x^2$ concave down?

1

- (A) $x < -\frac{1}{3}$
 (B) $x > -\frac{1}{3}$
 (C) $x < -3$
 (D) $x > 3$

5 The graph of the derivative of $y = f(x)$ is shown:



A maximum turning point of $y = f(x)$ occurs at:

1

- (A) $x = -1$
 (B) $x = 2$
 (C) $x = 5$
 (D) $x = 7$

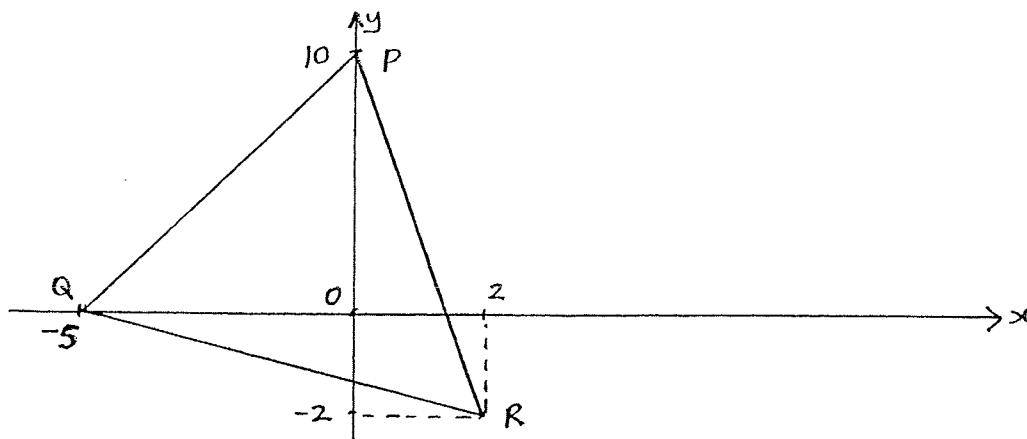
Section II: Short Answer

Question 6 (5 marks)

Commence a NEW page

Mark

In the following diagram, $\triangle PQR$ has vertices $P(0,10)$, $Q(-5,0)$, and $R(2,-2)$.



- | | | |
|-------|---|---|
| (i) | Show that PQ has length $5\sqrt{5}$ units. | 1 |
| (ii) | Show that PQ has equation $2x - y + 10 = 0$. | 1 |
| (iii) | Show that the perpendicular distance from R to PQ is $\frac{16}{\sqrt{5}}$ units. | 1 |
| (iv) | Find the coordinates of point S such that $PQRS$ is a parallelogram. | 1 |
| (v) | Find the area of parallelogram $PQRS$. | 1 |

Question 7 (8 marks)

Commence a NEW page

Mark

- | | | |
|-----|---|---|
| (a) | Point P divides the interval AB externally in the ratio $1:3$.
If A has coordinates $(-1,1)$ and B is $(3,-1)$, what are P 's coordinates? | 2 |
| (b) | The acute angle between the lines $3x - y + 7 = 0$ and $kx - y + 1 = 0$ is 45° .
Find the possible values of k . | 2 |
| (c) | Sketch the region where the inequalities $x + y < 2$ and $y \leq \sqrt{4 - x^2}$ both hold true. | 4 |

Question 8 (8 marks) Commence a NEW page **Mark**

Differentiate the following with respect to x .

(i) $y = \frac{x^2}{2\sqrt{x}}$ 2

(ii) $y = \frac{1+x}{1-x}$ 2

(iii) $y = (3x+4)^3$ 2

(iv) $y = \sqrt{5-2x}$ 2

Question 9 (8 marks) Commence a NEW page **Mark**

Consider the curve $y = 2x^3 + 3x^2 - 36x + 4$.

(i) Find the stationary points and determine their nature. 4

(ii) Find any points of inflexion. 2

(iii) Sketch the curve (x -intercepts not needed). 2

Question 10 (8 marks) Commence a NEW page **Mark**

(a) The line $2x + my + 1 = 0$ is tangent to the circle $x^2 + y^2 = \frac{1}{9}$.

Find the possible values of m . 3

(b) For what values of x is the function $y = \frac{1}{4}x^4 - 2x^2 + 3$ an increasing function? 2

(c) Find the equation of the tangent to $y = x^3 - 2x$ at the point $(2, 4)$. 2

(d) Evaluate the following limit: $\lim_{x \rightarrow 10} \frac{x^2 - 100}{x - 10}$ 1

Question 11 (8 marks)

Commence a NEW page

Mark

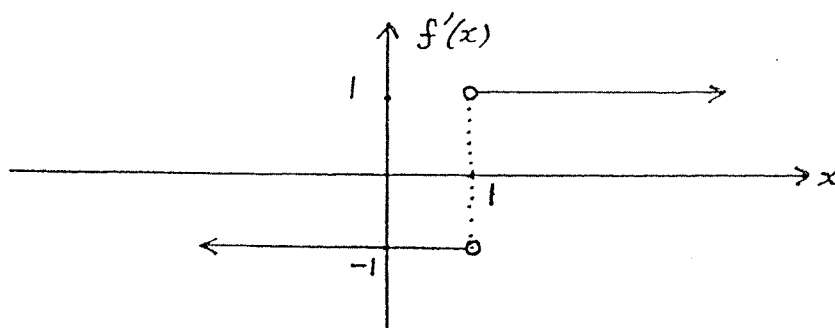
(a) The curve $y = ax^2 + bx + 2$ has a minimum turning point at $(-1, 1)$.

Evaluate a and b .

3

(b) Sketch a possible function which has the gradient function shown below:

2



(c) Differentiate from first principles the function $f(x) = (x+1)^2$.

3

END OF EXAM

$$\underline{Q1} \quad \frac{5-m}{-2-0} = \frac{m+3}{0-4}$$

$$-2m-6 = -20+4m$$

$$m = \frac{14}{6} = \frac{7}{3} \quad \therefore$$

(B)

✓

$$\underline{Q2} \quad \lim_{x \rightarrow \infty} \frac{3x - 2x^2}{5x^2 + 2x - 1}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3}{x} - 2}{5 + \frac{1}{x} - \frac{1}{x^2}} = -\frac{2}{5} \quad \therefore$$

(C)

✓

$$\underline{Q3} \quad \text{Test } P(\frac{1}{2}, 1) \rightarrow \left. \begin{array}{l} y \leq \frac{1}{x} \checkmark \\ x < y \checkmark \end{array} \right\} \therefore \rightarrow$$

(B)

✓

$$\text{Test } Q(-2, -1) \rightarrow \left. \begin{array}{l} y \leq \frac{1}{x} \checkmark \\ x < y \checkmark \end{array} \right\} \therefore \nearrow$$

$$\underline{Q4} \quad f'(x) = 3x^2 + 2x$$

$$f''(x) = 6x + 2$$

$$f'' < 0 \therefore 6x + 2 < 0 \therefore x < -\frac{1}{3} \therefore$$

(A)

✓

$$\underline{Q5} \quad x = 5 \quad \therefore$$

(C)

✓

Q6

$$(i) \quad PQ^2 = 5^2 + 10^2$$

$$= 125$$

$$\therefore PQ = \sqrt{125} = 5\sqrt{5}$$

✓

$$(ii) \quad m = \frac{10}{5} = 2$$

$$y\text{-int.} = 10$$

$$\therefore y = 2x + 10$$

$$\text{ie } 2x - y + 10 = 0.$$

(or alternative methods).

✓

$$(iii) \quad d = \frac{|2(2) + (-1)(-2) + 10|}{\sqrt{2^2 + (-1)^2}}$$

$$= \frac{|4 + 2 + 10|}{\sqrt{4 + 1}} = \frac{16}{\sqrt{5}}$$

✓

(iv) S is to R as P is to Q,

$$\therefore S = (2+5, -2+10) = (7, 8)$$

✓

$$(v) \quad A_{PQRS} = PQ \times d$$

$$= 5\sqrt{5} \times \frac{16}{\sqrt{5}}$$

$$= 80 \text{ units}^2.$$

✓

/ 5

Q7 (a)

$$A(-1, 1) \quad B(3, -1)$$

$$1 \quad : \quad -3$$

$$P = \left[\frac{(-3)(-1) + 1(3)}{1 + (-3)}, \frac{(-3)(1) + 1(-1)}{1 + (-3)} \right]$$

$$= \left(\frac{6}{-2}, \frac{-4}{-2} \right)$$

$$\therefore P = (-3, 2)$$

✓✓

$$(b) \quad m_1 = 3, \quad m_2 = k$$

$$\tan 45^\circ = \left| \frac{3 - k}{1 + 3k} \right|$$

✓

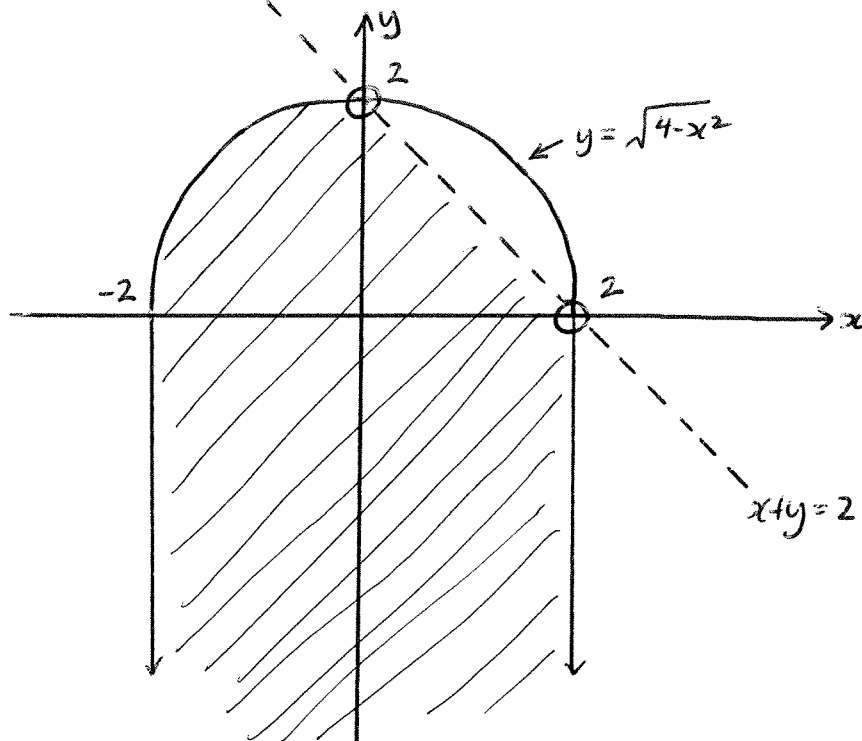
$$\therefore \left| \frac{3 - k}{1 + 3k} \right| = 1$$

$$\therefore \frac{3 - k}{1 + 3k} = 1 \quad \text{or} \quad \frac{3 - k}{1 + 3k} = -1$$

$$\therefore k = \frac{1}{2} \text{ or } -2.$$

✓

(c)



✓ correct semicircle

✓ correct dotted line

✓ correct region

✓ open circles

8

Q8

$$(i) \quad y = \frac{x^2}{2\sqrt{x}}$$

$$= \frac{1}{2} \cdot x^{\frac{3}{2}}$$

$$\therefore y' = \frac{1}{2} \cdot \frac{3}{2} \cdot x^{\frac{1}{2}}$$

$$\therefore y' = \frac{3}{4} x^{\frac{1}{2}} \quad \left(\text{or } \frac{3\sqrt{x}}{4} \right).$$

✓✓

$$(ii) \quad y = \frac{1+x}{1-x}$$

$$\therefore y' = \frac{(1-x) \cdot 1 - (1+x) \cdot -1}{(1-x)^2}$$

$$= \frac{1-x+1+x}{(1-x)^2}$$

$$\therefore y' = \frac{2}{(1-x)^2}$$

✓✓

$$(iii) \quad y = (3x+4)^3$$

$$\therefore y' = 3(3x+4)^2 \cdot 3$$

$$\therefore y' = 9(3x+4)^2$$

✓✓

$$(iv) \quad y = \sqrt{5-2x}$$

$$= (5-2x)^{\frac{1}{2}}$$

$$\therefore y' = \frac{1}{2} \cdot (5-2x)^{-\frac{1}{2}} \cdot -2$$

$$\therefore y' = - (5-2x)^{-\frac{1}{2}}$$

$$= \frac{-1}{\sqrt{5-2x}}$$

✓✓

Q9 $y = 2x^3 + 3x^2 - 36x + 4$

(i) $y' = 6x^2 + 6x - 36$
 $= 6(x^2 + x - 6)$
 $= 6(x+3)(x-2)$

$y'' = 12x + 6$
 $= 6(2x+1)$

Stat. points when $y' = 0$

$\therefore x = -3$
 $y = 85$

or $x = 2$
 $y = -40$

At $(-3, 85)$, $y'' = 6(2x-3+1) < 0$

$\therefore (-3, 85)$ is a maximum turning point.

At $(2, -40)$, $y'' = 6(4+1) > 0$

$\therefore (2, -40)$ is a minimum turning point.

(ii)

For inflexions, $y'' = 0$

$\therefore 6(2x+1) = 0$

$x = -\frac{1}{2}$

$y = 22\frac{1}{2}$

Test:

x	-1	$-\frac{1}{2}$	0
y''	$6(-1)$	0	$6(1)$

Change of concavity $\therefore (-\frac{1}{2}, 22\frac{1}{2})$ is an inflexion.

✓ correct y'

✓ correct x coords.

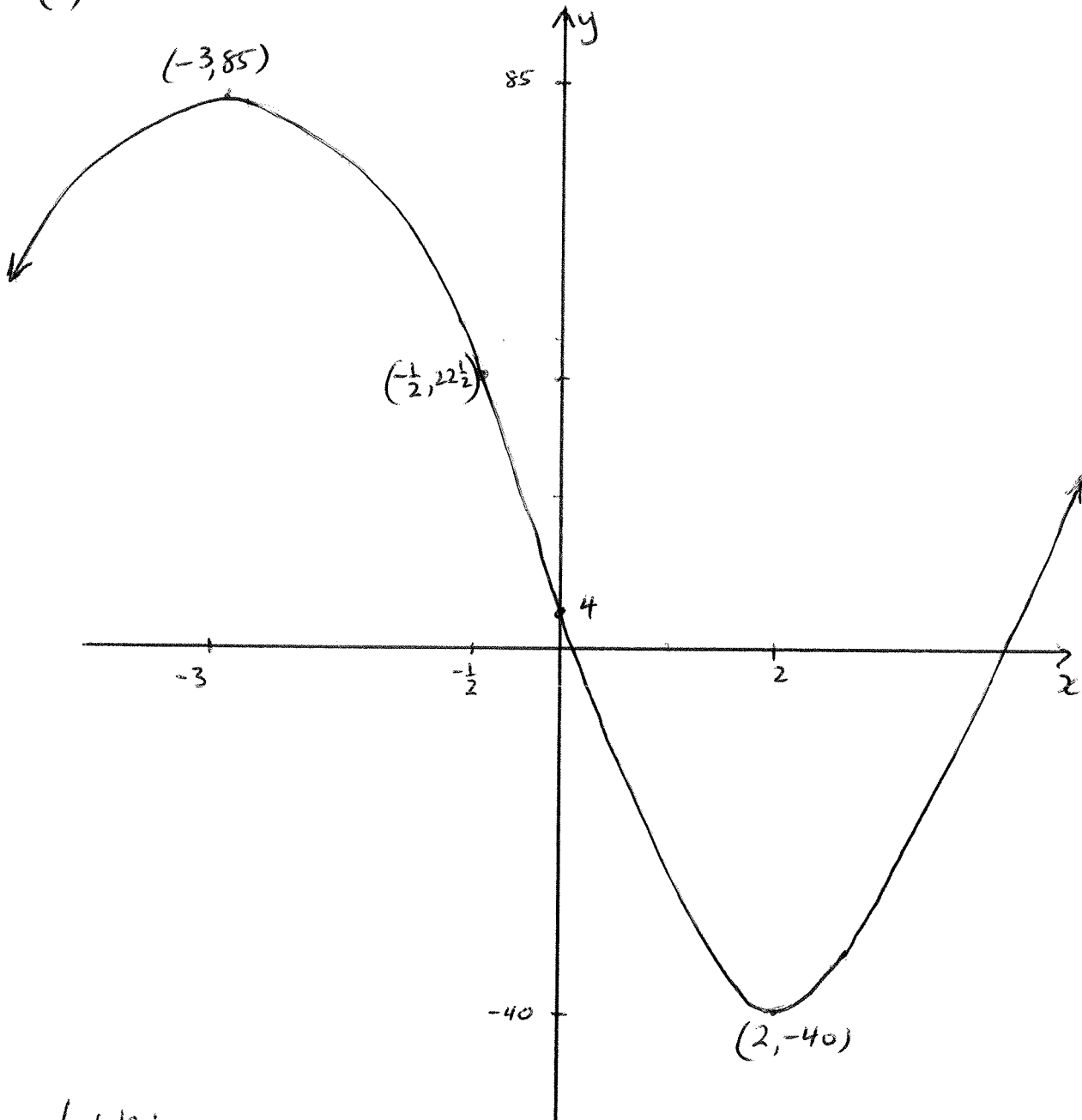
✓ correct y coords.

✓ Tests adequately

✓ correct inflexion

✓ Tests adequately

...[Q9-ctd.]
(iii)



(Note:
y-int. = 4)

✓✓

8

Q10

$$(a) \begin{cases} \text{Centre of circle} = (0,0) \\ \text{radius} = \frac{1}{3} \end{cases}$$

distance of line from $(0,0)$ is:

$$d = \frac{|2(0) + m(0) + 1|}{\sqrt{2^2 + m^2}} = \frac{1}{\sqrt{4+m^2}}$$

For tangency, $d = \frac{1}{3}$

$$\therefore \sqrt{4+m^2} = 3$$

$$\therefore 4+m^2 = 9$$

$$\therefore m = \pm \sqrt{5}$$

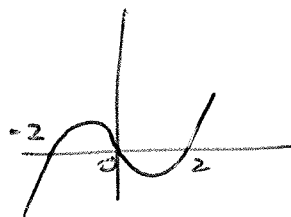
$$(b) \quad y = \frac{1}{4}x^4 - 2x^2 + 3 \text{ is increasing if } y' > 0$$

$$\therefore x^3 - 4x > 0$$

$$x(x^2 - 4) > 0$$

$$x(x-2)(x+2) > 0$$

$$\therefore -2 < x < 0 \text{ or } x > 2$$



$$(c) \quad y = x^3 - 2x \Rightarrow y' = 3x^2 - 2$$

$$\therefore m_T = 3(2)^2 - 2 \text{ at } (2,4) \quad \therefore m_T = 10$$

$$\therefore \text{tangent is } y - 4 = 10(x - 2)$$

$$[\text{i.e. } 10x - y - 16 = 0]$$

$$(d) \quad \lim_{x \rightarrow 10} \frac{x^2 - 100}{x - 10} = \lim_{x \rightarrow 10} \frac{(x-10)(x+10)}{x-10} = 20$$

Q11 (a) $y = ax^2 + bx + 2$, min. at $(-1, 1)$.

$y' = 2ax + b = 0$ for turning point.

$\therefore -2a + b = 0 \dots (1)$

Goes through $(-1, 1)$, so $1 = a(-1)^2 + b(-1) + 2$

$\therefore a - b = -1 \dots (2)$

$(1) + (2) \rightarrow -a = -1, \therefore a = 1$
 $b = 2$

✓✓✓

[ALT. The curve is clearly a parabola, with vertex $(-1, 1)$.

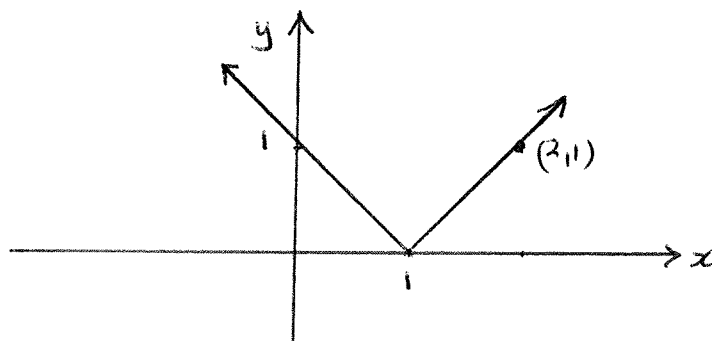
So it can be written $y = A(x+1)^2 + 1$

Since it goes through $(0, 2)$,

$\therefore 2 = A + 1 \therefore A = 1$

Thus $y = (x+1)^2 + 1 = x^2 + 2x + 2 \therefore a = 1$
 $b = 2$]

(b)



✓✓

(c) $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h+1)^2 - (x+1)^2}{h}$

$= \lim_{h \rightarrow 0} \frac{(x+h+1+x+1)(x+h+1-x-1)}{h}$

$= \lim_{h \rightarrow 0} \frac{(2x+h+2)(\cancel{h})}{\cancel{h}}$

$= 2x + 2$

✓

✓

✓

→ P.T.O.
for alternative

Q11 (c) ALTERNATIVELY ...

$$\begin{aligned}
 [1] f'(x) &= \lim_{u \rightarrow x} \frac{f(u) - f(x)}{u - x} \\
 &= \lim_{u \rightarrow x} \frac{(u+1)^2 - (x+1)^2}{u - x} \\
 &= \lim_{u \rightarrow x} \frac{(u+1+x+1)(u+1-x-1)}{u - x} \\
 &= \lim_{u \rightarrow x} \frac{(u+x+2)(u-x)}{u-x} \\
 &= 2x+2.
 \end{aligned}$$

$$\begin{aligned}
 \text{ALT. [2]} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h+1)^2 - (x+1)^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{[(x+1)+h]^2 - (x+1)^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{(x+1)^2} + 2(x+1)h + h^2 - \cancel{(x+1)^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2(x+1)h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} 2(x+1) + h \\
 &= 2(x+1).
 \end{aligned}$$